

Arithmetic Wavefront Set and Local Langlands Correspondence

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Classical Groups

Notations:

F : local field of characteristic 0, i.e. $F = \mathbb{R}, \mathbb{C}$ or a finite extension of p -adic fields $\mathbb{Q}_p$

$F(\delta)$: quadratic extension of F

$E := F$ or $F(\delta)$

(V, q) : n -dimensional linear space over E equipped with a non-degenerate ϵ -Hermitian form q , with q symmetric or symplectic when $E = F$, Hermitian or skew-Hermitian when $E = F(\delta)$.

$$\epsilon = \begin{cases} 1, & q \text{ symmetric or Hermitian} \\ -1, & q \text{ symplectic or skew-Hermitian} \end{cases} \quad (1)$$

$G_n := \text{Isom}(V, q)^\circ$, the identity component of $\text{Isom}(V, q)$, or the metaplectic group Mp_{2n} , where $n = \lfloor n/2 \rfloor$.

$\mathfrak{g}_n$: the Lie algebra of G_n

Classical Groups

Cases that we focus on:

- ① $G_n(F) \simeq U_n(F)$, corresponding to $E = F(\delta)$, with q being Hermitian or skew-Hermitian
- ② $G_n(F) \simeq SO_n(F)$, corresponding to $E = F$, with q being symmetric
- ③ $G_n(F) \simeq Sp_{2n}(F)$, corresponding to $E = F$, with q being symplectic
- ④ $G_n(F) \simeq Mp_{2n}(F)$, the unique nontrivial double cover of $Sp_{2n}(F)$

From now on, G_n denotes a classical group isomorphic to one of U_n , SO_n , Sp_{2n} , or the metaplectic group Mp_{2n} .

(You can think of easier cases where $G_n \simeq GL_n, SL_n$.)

Analytic Wavefront Set

(π, V_π) : an irreducible admissible representation of G_n .

Θ_π : distribution character of π , which is a distribution on G_n , defined by

$$\langle \Theta_\pi, f \rangle := \text{Tr}(\pi(f)), \quad f \in C_c^\infty(G_n), \quad (2)$$

where

$$\pi(f) := \int_{G_n} f(g) \pi(g) dg. \quad (3)$$

Consider the adjoint action of $G_n(F)$ on its Lie algebra $\mathfrak{g}_n(F)$.

$\mathcal{N}_F(\mathfrak{g}_n)_\circ$: the set of F -rational nilpotent orbits in $\mathfrak{g}_n(F)$

Analytic Wavefront Set

We have the asymptotic expansion of Θ_π near the identity

$$\Theta_\pi \circ \exp = \sum_{O \in \mathcal{N}_F(\mathfrak{g}_n)_o} c_O \widehat{\mu_O}, \quad (4)$$

where $c_O \in \mathbb{C}$, $\widehat{\mu_O}$ is the Fourier transform of the (normalized) measure on $O \subseteq \mathfrak{g}_n(F)$, and $\exp : \mathfrak{g}_n(F) \rightarrow G_n(F)$ is a diffeomorphism near 0.

Definition

Define the analytic wavefront set of the representation π to be

$$WF_{tr}(\pi) := \{O \in \mathcal{N}_F(\mathfrak{g}_n)_o : c_O \neq 0\}. \quad (5)$$

When $F = \mathbb{R}$, (5) coincides with the definition by Hörmander:

$$WF_{tr}(\pi) := WF(\Theta_\pi)$$

Algebraic Wavefront Set

(π, V_π) : an irreducible admissible representation of G_n .

Pick an F -rational orbit $O \in \mathcal{N}_F(\mathfrak{g}_n)_o$, and pick any $X \in O$.

$Wh_X(\pi)$: the space of (generalized) **Whittaker models** of π associated to X , which turns out only depending on the orbit O .

Definition

Define the algebraic wavefront set $WF_{wm}(\pi)$ of π to be

$$WF_{wm}(\pi) := \{O \in \mathcal{N}_F(\mathfrak{g}_n)_o : Wh_X(\pi) \neq 0, X \in O\} \quad (6)$$

Definition of $Wh_X(\pi)$:

Extend X to a $\mathfrak{sl}_2$ -triple $\{X, H, Y\}$, which gives the weight space decomposition

$$\mathfrak{g}_n = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_{n,i}^H \quad (7)$$

where

$$\mathfrak{g}_{n,i}^H(F) := \{x \in \mathfrak{g}_n(F) : ad(H)(x) = ix\}. \quad (8)$$

Let $u_X := \bigoplus_{i \leq -1} \mathfrak{g}_{n,i}^H$, $U_X := \exp(u_X) \subseteq G_n$.

Definition

Define the space of generalized Whittaker models of π associated to X as

$$Wh_X(\pi) := Hom_{U_X}(V_\pi, V_{\psi_X}) \quad (9)$$

where V_{ψ_X} is an irreducible representation of U_X , called the smooth oscillator representation associated to X .

Here ψ_X is a character on $U_{X,2} := \exp(\bigoplus_{i \leq -2} \mathfrak{g}_{n,i}^H) \subseteq U_X$, induced canonically from X .

If $\mathfrak{g}_{n,-1} = 0$, then $U_{X,2} = U_X$ and V_{ψ_X} is precisely ψ_X , and then $Wh_X(\pi) = Hom_{U_X}(\pi, \psi_X)$.

Conjecture

We have

$$WF_{tr}(\pi)^{max} = WF_{wm}(\pi)^{max} \quad (10)$$

Here the maximality is defined by the F -stable topological order:

$$O_1 \leq_{st} O_2 \iff O_1^{st} \subseteq \overline{O_2^{st}}, \quad (11)$$

where O_i^{st} is the F -stable ($\bar{F}$ -rational) orbit containing O_i .

Example:

$G = SL_2(\mathbb{R})$, $(\pi, V_\pi) = (D_k^+, \mathcal{H}_k)$ the homological discrete series representation of weight $k \in \mathbb{Z}_{\geq 2}$, where $\mathcal{H}_k \subseteq L^2(\mathbb{H}, y^{k-2} dx dy)$ consisting of holomorphic functions.

Then $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{R})$

Nilpotent orbits:

$$O^+ \ni \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad O^- \ni \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad 0 := \{0\}. \quad (12)$$

We have

$$WF_{wm}(D_k^+) = WF_{tr}(D_k^+) = \{O^+, 0\}. \quad (13)$$

Local Langlands Correspondence

$\Pi_F(G_n)$: the set of (equivalence classes of) irreducible admissible representations of $G_n^*(F)$.

$\Phi_F(G_n)$: the set of (equivalent classes of) **local L -parameters** of G_n .

Local Langlands correspondence (LLC) is a canonical map

$$\mathcal{R}_{F,G_n} : \Pi_F(G_n) \longrightarrow \Phi_F(G_n) \quad (14)$$

satisfying

- ① the **L -packet** $\Pi_\phi(G_n) := \mathcal{R}_{F,G_n}^{-1}(\phi)$ is non-empty and finite,
 $\forall \phi \in \Phi_F(G_n)$
- ② compatibility with L -functions
- ③ compatibility with ϵ -factors
- ④

Arithmetic Side: Local L -parameters

A local L -parameter ϕ of G_n is a homomorphism

$$\phi : WD_F \longrightarrow {}^L G_n := G_n^\vee \rtimes \text{Gal}(\bar{F}/F) \quad (15)$$

compatible with the action of the **Weil group** W_F or the Galois group on both sides. Here

$$WD_F := \begin{cases} W_F, & F \text{ archimedean} \\ W_F \times SL_2(\mathbb{C}), & F \text{ non-archimedean} \end{cases} \quad (16)$$

is the Weil-Deligne group over F , $G_n^\vee$ is the complex dual of G_n .

Arithmetic Side: Local L -parameters

Definition of W_F :

① F archimedean:

① $F = \mathbb{R}$: $W_{\mathbb{R}} := \mathbb{C}^{\times} \cup j\mathbb{C}^{\times}$, where $jzj = \bar{z}, \forall z \in \mathbb{C}^{\times}$

② $F = \mathbb{C}$: $W_{\mathbb{C}} := \mathbb{C}^{\times}$

② F non-archimedean: W_F is defined to be the dense subgroup of the topological group $\text{Gal}(\bar{F}/F)$ containing the inertia subgroup $\mathcal{I}_F$ as an open subgroup such that

$$1 \longrightarrow \mathcal{I}_F \longrightarrow W_F \longrightarrow \mathbb{Z} \longrightarrow 0 \quad (17)$$

is exact. (Recall the exact sequence

$$1 \longrightarrow \mathcal{I}_F \longrightarrow \text{Gal}(\bar{F}/F) \longrightarrow \widehat{\mathbb{Z}} \longrightarrow 0 \quad (18)$$

from algebraic number theory.)

Let G_n^* denote a classical group isomorphic to one of U_n, SO_n, Sp_{2n} or the metaplectic group Mp_{2n} .

$\Phi_{gen}(G_n^*)$: the set of **generic** local L -parameters of G_n^* .

Pick $\phi \in \Phi_{gen}(G_n^*)$, define the component group

$$\mathcal{S}_\phi := S_\phi / S_\phi^\circ \quad (19)$$

where $S_\phi := \text{Cent}_{G_n^*}(\text{Im } \phi)$.

The **Vogan L -packet** $\Pi_\phi[G_n^*] := \bigcup_{G_n} \Pi_\phi(G_n)$, a disjoint union where G_n runs through all pure inner forms of G_n^* .

The set of Whittaker data

$$\mathcal{Z} := \begin{cases} F^\times / (F^\times)^2, & E = F \\ F^\times / N_{E/F}(F^\times), & E = F(\delta) \end{cases} \quad (20)$$

Genericity:

Definition

Let $\pi \in \Pi_F(G_n)$, $\mathcal{O} \in \mathcal{N}_F(\mathfrak{g}_n)_\circ$. Say π is generic with respect to $\mathcal{O}$ if there exists $X \in \mathcal{O}$ such that $\text{Wh}_X(\pi) \neq 0$.

Definition

Suppose $\phi \in \Phi_F(G_n^*)$ is a local L -parameter of G_n^* .

- ① If $G_n^* \neq \text{Mp}_{2n}$, say ϕ is generic if $\Pi_\phi[G_n^*]$ contains a generic member with respect to some Whittaker datum $a \in \mathcal{Z}$;
- ② If $G_n^* = \text{Mp}_{2n}$, say ϕ is generic if it is generic as a local L -parameter of SO_{2n+1} .

(Note that $(\text{Mp}_{2n})^\vee = (\text{SO}_{2n+1})^\vee = \text{Sp}_{2n}(\mathbb{C})$.)

Vogan version of local Langlands correspondence:

Conjecture

There exists a canonical bijection

$$\iota_a : \Pi_\phi[G_n^*] \longrightarrow \widehat{\mathcal{S}}_\phi \quad (21)$$

for any $\phi \in \Phi_{\text{gen}}(G_n^)$, depending on the choice of the Whittaker datum $a \in \mathcal{Z}$.*

Fix a generic local L -parameter ϕ and $a \in \mathcal{Z}$.

Assuming (21), for $\pi \in \Pi_\phi[G_n^*]$, there exists a unique $\chi \in \widehat{\mathcal{S}}_\phi$ such that $\iota_a(\pi) = \chi$. Call (ϕ, χ) the **enhanced L -parameter** of G_n^* corresponding to π .

Conversely, we can denote by $\pi = \pi_a(\phi, \chi)$ the representation corresponding to (ϕ, χ) .

L-parameter Descent

Given an enhanced L -parameter (ϕ, χ) of G_n^* , by Gan-Gross-Prasad theorem, for $0 < l \leq n$ we consider relevant pairs of classical groups $(G_n^*, H_{[\frac{n-l}{2}]}^*)$:

- ① (SO_n, SO_{n-l}) , if $E = F, \epsilon = 1$
- ② (Sp_{2n}, Mp_{2n-l}) or (Mp_{2n}, Sp_{2n-l}) , if $E = F, \epsilon = -1$
- ③ (U_n, U_{n-l}) , if $E = F(\delta)$

where we require $(-1)^{l-1} = \epsilon$ if $E = F$.

L-parameter Descent

Definition

Given (ϕ, χ) , l as above, for every $z \in \mathcal{Z}$, define the l -th descent of (ϕ, χ) along z to be

$$\begin{aligned} \mathfrak{D}_l^z(\phi, \chi) &:= \{(\widehat{\varphi}, \widehat{\chi}') : \varphi \in \Phi_{\text{gen}}(H_{[\frac{n-1}{2}]}^*) \text{ with opposite type to } \phi \\ &\quad \text{such that } \chi_{\phi, \varphi}^z = \chi, \\ &\quad \chi' = \chi_{\varphi, \phi}^z \in \widehat{S}_{\varphi}\} \end{aligned}$$

We define the l -th descent of (ϕ, χ) to be

$$\mathfrak{D}_l(\phi, \chi) := \bigcup_{z \in \mathcal{Z}} \mathfrak{D}_l^z(\phi, \chi). \quad (22)$$

Arithmetic Wavefront Set

Suppose $\pi \longleftrightarrow (\phi, \chi)$ via Vogan local Langlands correspondence ι_a .
(i.e. $\pi = \pi_a(\phi, \chi)$)

Arithmetic wavefront set $WF_{ari}(\pi)$:

collecting arithmetic data of (ϕ, χ) by looking at the L -parameter descents of (ϕ, χ) .

Partitions $\underline{p} := [p_1^{m_1}, \dots, p_r^{m_r}]$ of n .

The set $\mathcal{N}_F(\mathfrak{g}_n)_\circ^{st}$ of F -stable nilpotent orbits.

Partitions and F -stable Nilpotent Orbits

Proposition

There exists a canonical bijection from $\mathcal{N}_F(\mathfrak{g}_n)_\circ^{st}$ to the subset $\mathcal{P}(V, q)$ of all partitions $\underline{p}$ of n , except that when $G_n = SO_{2n}$, a totally even partition corresponds to two F -stable nilpotent orbits.

Call a partition $\underline{p} = [p_1^{m_1}, \dots, p_r^{m_r}]$ **totally even**, if m_i, p_i are even for all i .
 $\underline{p} \in \mathcal{P}(V, q)$:

- 1 If $G_n \simeq Sp_{2n}$ or Mp_{2n} , m_i is even for every odd p_i .
- 2 If $G_n \simeq SO_n$, m_i is even for all even p_i .

Young Tableaux and F -rational Nilpotent Orbits

Suppose $G_n \simeq \text{Isom}(V, q)^\circ$.

(V, q) -admissible sesquilinear Young tableaux:

$$\left(\underline{p}, \{(U_{(p_i)}^h, q_{(p_i)}^h)\}_{i=1}^r \right). \quad (23)$$

$\mathcal{Y}(V, q)$: the set of (equivalence classes of) (V, q) -admissible sesquilinear Young tableaux.

Proposition

There exists a canonical bijection $\mathfrak{C}$ from $\mathcal{N}_F(\mathfrak{g}_n)_\circ$ to $\mathcal{Y}(V, q)$, except that when $G_n = \text{SO}_{2n}$, an (V, q) -admissible sesquilinear Young tableau may correspond to two F -rational nilpotent orbits.

Arithmetic Wavefront Set

Suppose $\pi = \pi_a(\phi, \chi) \in \Pi_\phi[G_n^*]$ with ϕ generic, $\chi \in \widehat{\mathcal{S}_\phi}$, determining a pure inner form $G_n \simeq \text{Isom}(V, q)^\circ$ of G_n^* such that $\pi \in \Pi_\phi(G_n)$.

$\mathcal{T}_a(\phi, \chi)$: the set of pre-tableaux associated to (ϕ, χ) via the Whittaker datum a .

$\mathcal{L}_a(\phi, \chi) \subseteq \mathcal{T}_a(\phi, \chi)$: the set of **L -descent pre-tableaux** associated to (ϕ, χ) via a , consisting of all decreasing pre-tableaux in $\mathcal{T}_a(\phi, \chi)$.

$\mathcal{Y}_a(\phi, \chi) \subseteq \mathcal{Y}(V, q)$: the subset of (V, q) -admissible sesquilinear Young tableaux constructed canonically from $\mathcal{L}_a(\phi, \chi)$.

Definition

By the bijection $\mathfrak{C} : \mathcal{N}_F(\mathfrak{g}_n)_\circ \longrightarrow \mathcal{Y}(V, q)$, we define the arithmetic wavefront set as

$$WF_{\text{ari}}(\pi) := WF_a(\phi, \chi) := \mathfrak{C}^{-1}(\mathcal{Y}_a(\phi, \chi)) \quad (24)$$

Wavefront Set Conjecture

Conjecture

Let F be a local field of characteristic 0, $G_n^* = \text{Isom}(V, q)^\circ$ or $\text{Mp}_{2n}(F)$. Let ϕ be a generic local L -parameter of G_n^* . For any $\pi \in \Pi_\phi[G_n^*]$, say, $\pi \in \Pi_\phi(G_n)$, we have

$$\text{WF}_{\text{wm}}(\pi)^{\max} = \text{WF}_{\text{ari}}(\pi)^{\max} = \text{WF}_{\text{tr}}(\pi)^{\max} \quad (25)$$

where the maximality are defined under F -stable topological order in $N_F(\mathfrak{g}_n)_\circ$, where $\mathfrak{g}_n$ is the Lie algebra of G_n .

Example:

$G = \text{SL}_2(\mathbb{R})$, $(\pi, V_\pi) = (D_k^+, \mathcal{H}_k)$. We have

$$\text{WF}_{\text{ari}}(D_k^+)^{\max} = \{O^+\} = \text{WF}_{\text{wm}}(D_k^+)^{\max} = \text{WF}_{\text{tr}}(O^+)^{\max}. \quad (26)$$

When $F = \mathbb{R}$:

Theorem

Let ϕ be a generic local L -parameter of G_n^* . Then

- 1 if $\widehat{\mathcal{S}}_\phi$ is trivial, then $WF_{ari}(\pi)^{max}$ consists of all regular F -rational nilpotent orbits;
- 2 if $\widehat{\mathcal{S}}_\phi$ is nontrivial, then

$$WF_{ari}(\pi)^{max} = \{O(\pi)\} \quad (27)$$

contains precisely one F -rational nilpotent orbit.

(WF_{ari} can be computed explicitly.)

References

1. Dihua Jiang and Dongwen Liu and Lei Zhang, *Arithmetic wavefront sets and generic L-packets*, *Advances in Mathematics*, **483**(2025), 110668.
2. Bushnell, Colin J., and Guy Henniart, *The Local Langlands Conjecture for $GL(2)$* , Springer-Verlag Berlin Heidelberg, 2006.
3. W.T. Gan, B. Gross, D. Prasad, *Symplectic local root numbers, central critical L-values, and restriction problems in the representation theory of classical groups*, in: *Sur les conjectures de Gross et Prasad. I*, *Astérisque* 346 (2012) 1–109.
4. H. Atobe, W.T. Gan, *On the local Langlands correspondence and Arthur conjecture for even orthogonal groups*, *Represent. Theory* 21 (2017) 354–415.
5. R. Langlands, *On the classification of irreducible representations of real algebraic groups*, in: *Representation Theory and Harmonic Analysis on Semisimple Lie Groups*, in: *Math. Surveys Monogr.*, vol. 31, Amer. Math. Soc., Providence, RI, 1989, pp. 101–170.

6. T. Kaletha, *Representations of Reductive Groups over Local Fields*, EMS Press, Berlin, 2023, pp. 2948–2975.
7. J. Arthur, *The Endoscopic Classification of Representations. Orthogonal and Symplectic Groups*, American Mathematical Society Colloquium Publications, vol. 61, American Mathematical Society, Providence, RI, 2013.
8. C. Mok, *Endoscopic classification of representations of quasi-split unitary groups*, Mem. Am. Math. Soc. 235 (2015) 1108.
9. R. Gomez, C.-B. Zhu, *Local theta lifting of generalized Whittaker models associated to nilpotent orbits*, Geom. Funct. Anal. 24 (3) (2014) 796–853.
10. D. Jiang, L. Zhang, *Local root numbers and spectrum of the local descents for orthogonal groups: p -adic case*, Algebra Number Theory 12 (6) (2018) 1489–1535.

11. R. Howe, *Wave front sets of representations of Lie groups*, in: Automorphic Forms, Representation Theory and Arithmetic, Bombay, 1979, in: Tata Inst. Fund. Res. Studies in Math., vol. 10, Tata Institute of Fundamental Research, Bombay, 1981, pp. 117–140.
12. C. Mœglin, J.-L. Waldspurger, *Modèles de Whittaker dégénérés pour des groupes p -adiques*, Math. Z. 196 (3) (1987) 427–452.
13. W.T. Gan, G. Savin, *Representations of metaplectic groups I: epsilon dichotomy and local Langlands correspondence*, Compos. Math. 148 (6) (2012) 1655–1694.
14. H. Matumoto, *C - ∞ -Whittaker vectors corresponding to a principal nilpotent orbit of a real reductive linear Lie group, and wave front sets*, Compos. Math. 82 (2) (1992) 189–244.
15. J. Adams, D. Vogan, *Associated varieties for real reductive groups*, Pure Appl. Math. Q. 17 (4) (2021) 1191–1267.

Thank you!